

ABHINAV YELLAMELLI

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Education

Indian Institute of Technology, Madras

M.Tech, Electric Vehicles

Ongoing

CGPA: -

Visvesvaraya National Institute of Technology (VNIT), Nagpur

Bachelor of Technology in Electrical and Electronics Engineering

2020 – 2024

CGPA: 6.87/10

Experience

Graduate Engineer Trainee (GET), Jindal Stainless Ltd, Hisar, Haryana

Aug 2024 – May 2025

- Performed preventive and breakdown maintenance of electrical and instrumentation systems in the Steckel Hot Rolling Mill, ensuring high equipment availability.
- Executed engineering projects including IP camera installation, **MCC** replacement, **auto burner ignition system** implementation, and shot blasting drive modernization.
- Participated in the conversion of four reheating furnaces from furnace oil to natural gas, improving operational efficiency and reducing fuel costs.
- Performed calibration and maintenance of field instruments including **RTDs**, **thermocouples**, **pressure switches**, and **transmitters**, while preparing calibration charts and documentation as per plant quality procedures.
- Conducted SAP-based asset audits by verifying instrumentation equipment in the field, ensuring assets were correctly installed, identified, and matched with departmental records.

Relevant Coursework

Power Converter Design for EVs, Power Electronics & Motor Drives, Foundations to EV Technology, Fundamentals of Automotive Systems, Energy Storage Systems, Electric Drives & Their Control, Power Electronics, Advanced Power Electronics, Industrial Automation, MicroControllers, Smart Grid, Renewable Energy Systems.

Projects

Project at VNIT Nagpur: Battery Management System for Microgrid.

- Conducted a comprehensive literature review on Battery Energy Storage Systems (BESS) and evaluated different battery chemistries, selecting **Li-ion** batteries based on energy density, efficiency, lifecycle, and safety characteristics.
- Implemented and compared **Coulomb Counting** and **Kalman Filter** algorithms for **State of Charge (SOC)** estimation using MATLAB/Simulink.
- Designed and evaluated **passive** and **active cell-balancing techniques**, demonstrating the advantages of active balancing in minimizing energy losses and improving battery efficiency.
- Simulated and compared **switched-capacitor** and **switched-inductor** active-balancing topologies, identifying switched-capacitor balancing as the more effective approach under the simulated conditions.
- Investigated an **inductor-based Buck-Boost** cell-balancing topology from recent research literature and attempted its implementation in **MATLAB/Simulink**, identifying simulation complexity and computational limitations during model development.

Micro-Controller(EEL330) Course Project: One Touch Women's Safety Alarm (Arduino IDE)

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- Developed an **Arduino Nano**-based emergency alert device using **GSM** and **GPS Modules** that transmits the device's location to a pre-programmed SOS mobile number when a push button is activated in an emergency.
- Implemented wireless communication using a **433 MHz RF module** and developed the application in **Embedded C++** using **Arduino IDE**.

Technical Skills

Languages: C, Python.

Simulation Tools: MATLAB/Simulink, MULTISIM.